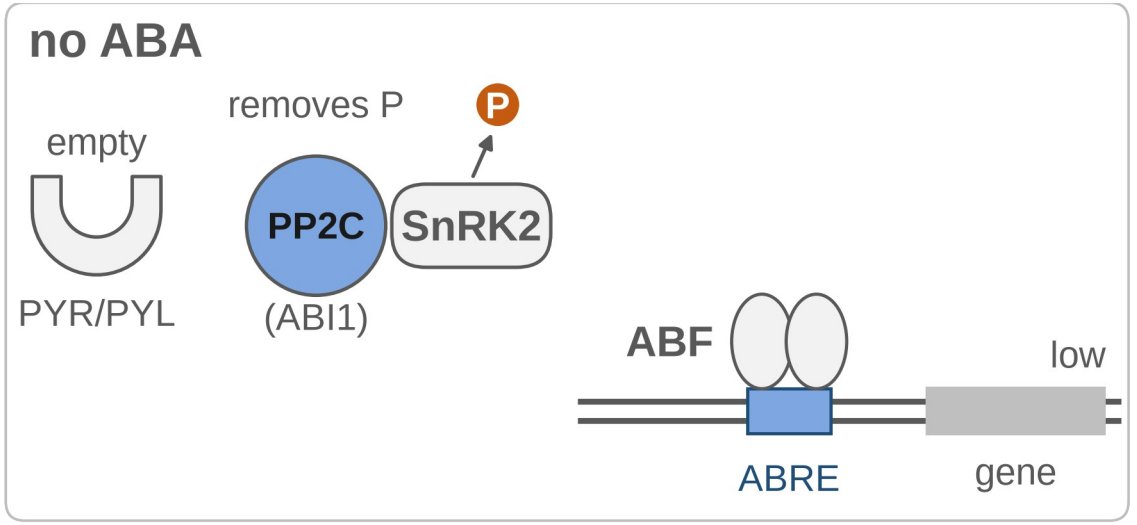
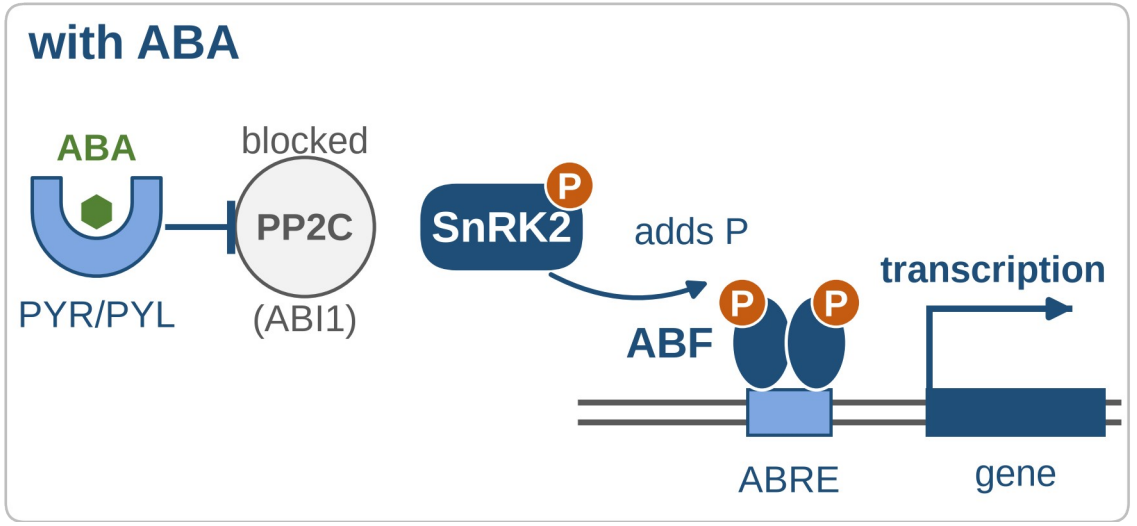
# The biology and the experiment

# ABA switches on stress genes

- Drought, salt, cold → plants make **abscisic acid (ABA)**
- ABA receptor blocks the phosphatase **ABI1** (PP2C)
- Kinases (**SnRK2**) become active
- They phosphorylate **ABF** transcription factors
- ABFs bind the **ABRE** in promoters → genes on

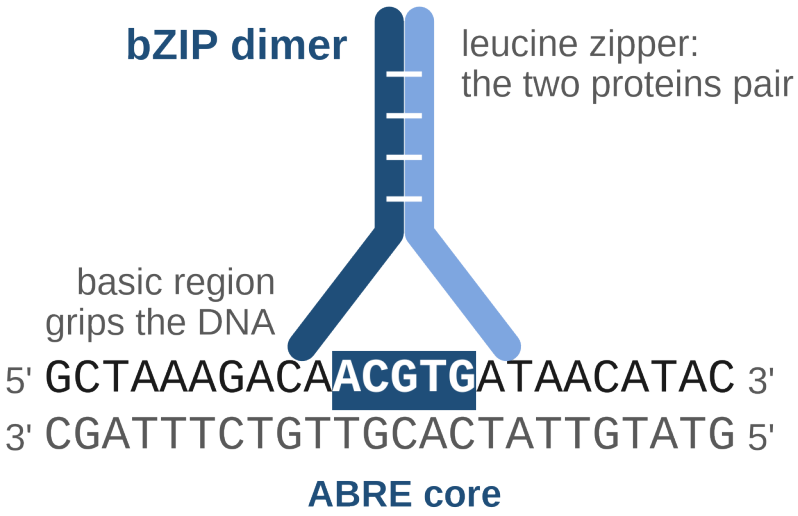



Cutler et al. 2010, Annu Rev Plant Biol 61:651; Fujii et al. 2009, Nature 462:660

# ABF3 binds the ABRE as a dimer

- ABFs are **bZIP** proteins: a basic DNA-binding region + a leucine zipper
- Two bZIPs pair (dimer) and grip the DNA
- **ABRE**: a short motif with the core ACGTG
- ABF1–4 have very similar DNA-binding regions




**the same core in five promoters (toy)**

| | | |
|---|---|---|
| geneA | ACG**ACGTG**ACGAAACTTGTTGGCCCAGT | + strand |
| geneB | GTGAATCGCTTAAGGGT**CACGT**AAGTGT | − strand |
| geneC | ACCCCATCGGA**ACGTG**ATTTTTATTACA | + strand |
| geneD | CTCAGAAACAGAACTCGGGTA**ACGTG**GA | + strand |
| geneE | AGTGCG**CACGT**ACTCGCTATGAATCTCT | − strand |

CACGT on the + strand = ACGTG read on the − strand

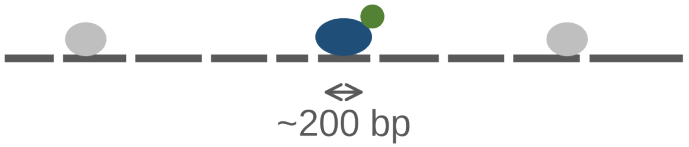
Fujita et al. 2011, J Plant Res 124:509; Yoshida et al. 2010, Plant J 61:672

# ChIP-seq: pull down, sequence, map

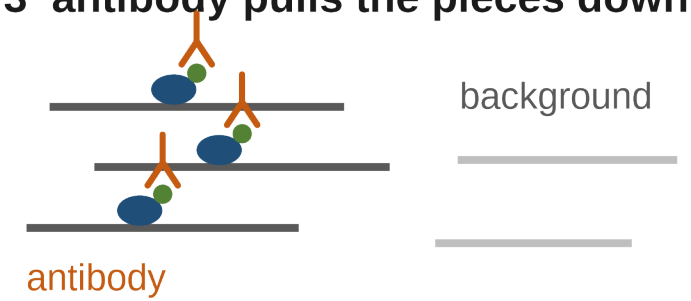
**1 tagged protein on its site**



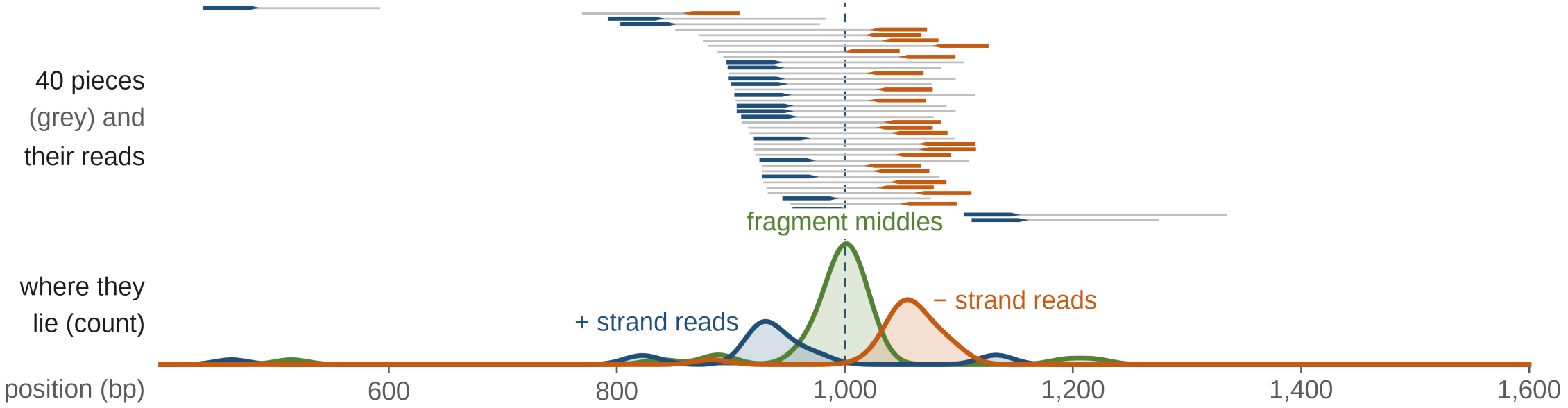

**2 cross-link, cut to ~200-bp pieces**




**3 antibody pulls the pieces down**

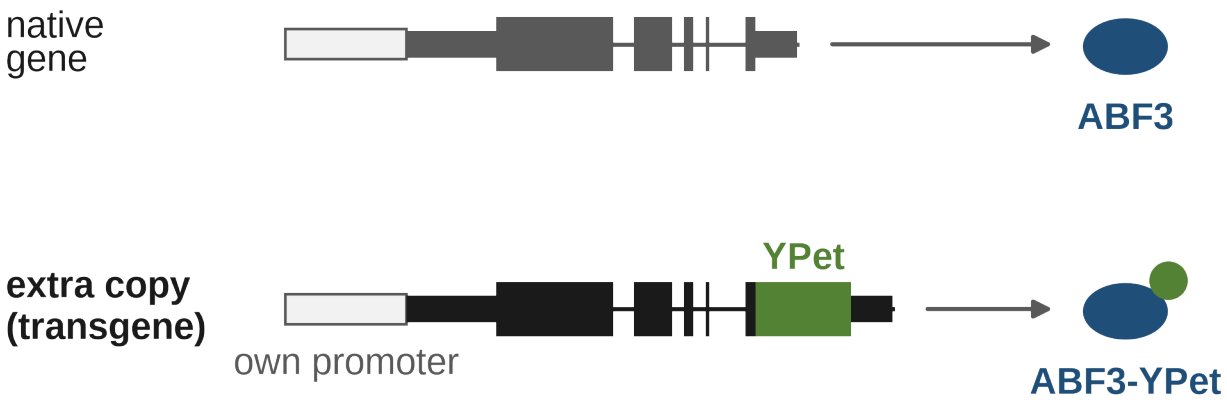



**4 sequence 50 bp from one end of each piece, map to the genome**



# ABF3-YPet: a tag instead of an antibody

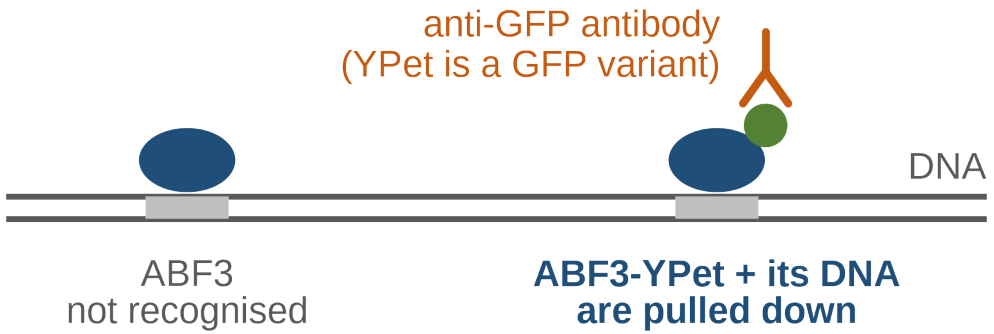
## the ABF3-YPet plant has two copies of ABF3




same promoter: both proteins are made
in the same cells at the same time

why a tag? good antibodies against single plant TFs are rare;
with a tag, one antibody works for every TF (21 in the study)

## ChIP with an anti-GFP antibody

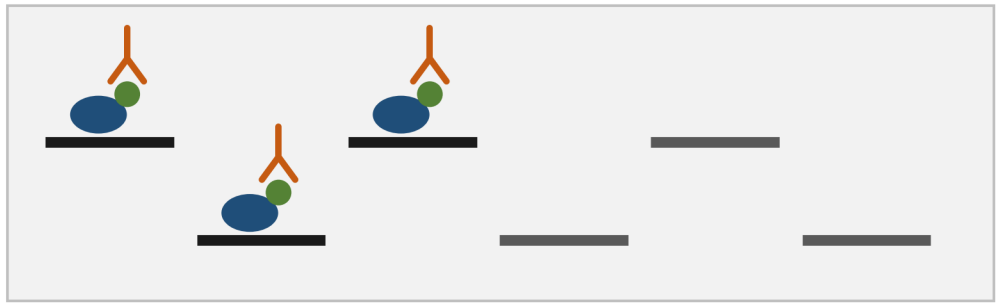



the ChIP finds where the tagged ABF3 sits;
we assume the tag does not change where ABF3 binds

Song et al. 2016, Science 354:aag1550

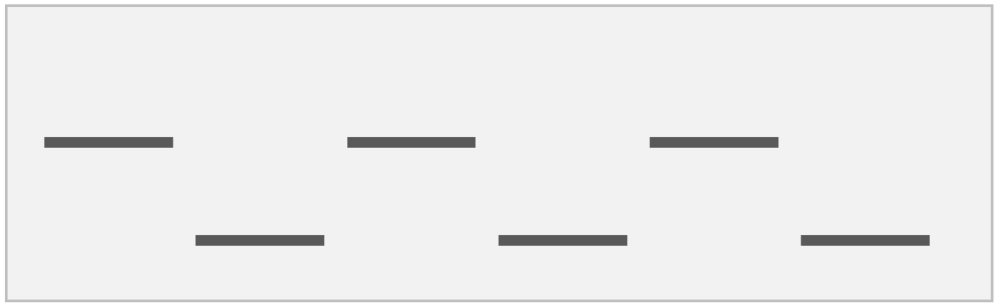
# The control: same ChIP, nothing to catch

**ABF3-YPet plants + anti-GFP**



**pulled down: ABF3 sites + background**

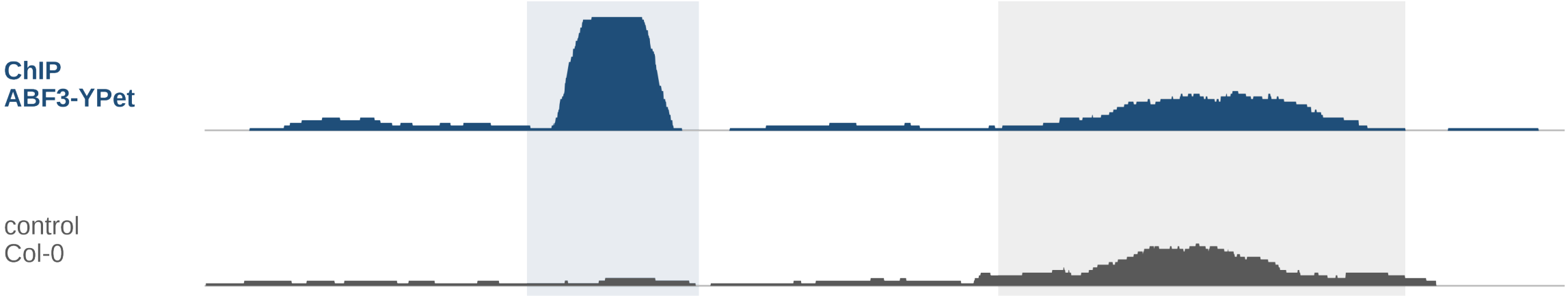
**Col-0 (no YPet) + anti-GFP**



pulled down: background only

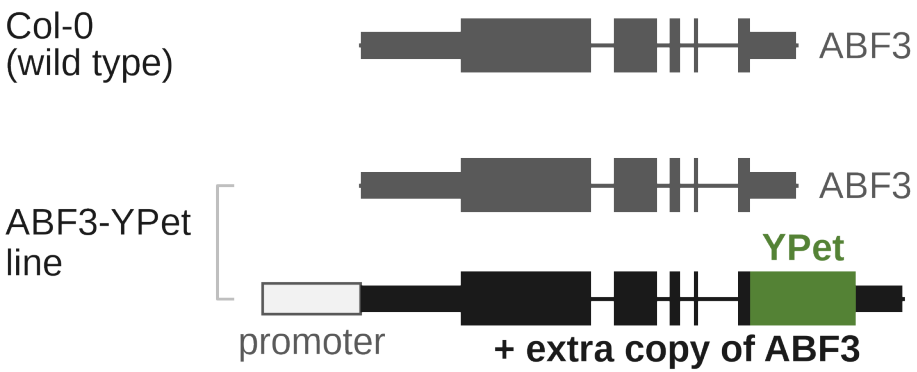
**binding site: ChIP >> control**

background: high in both - not a binding site

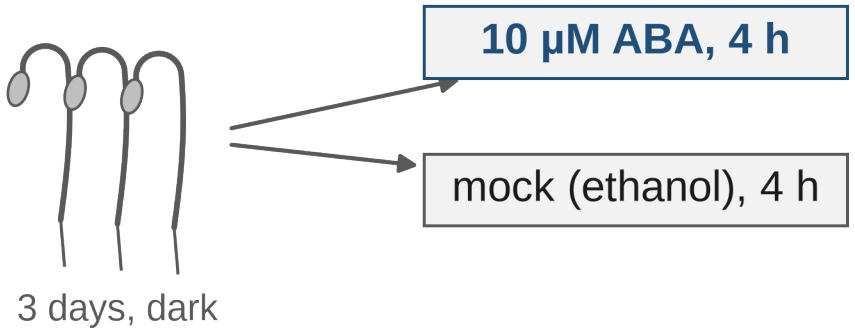


# Treatment and samples

## plants




## treatment




ChIP: anti-GFP antibody (it also binds YPet)

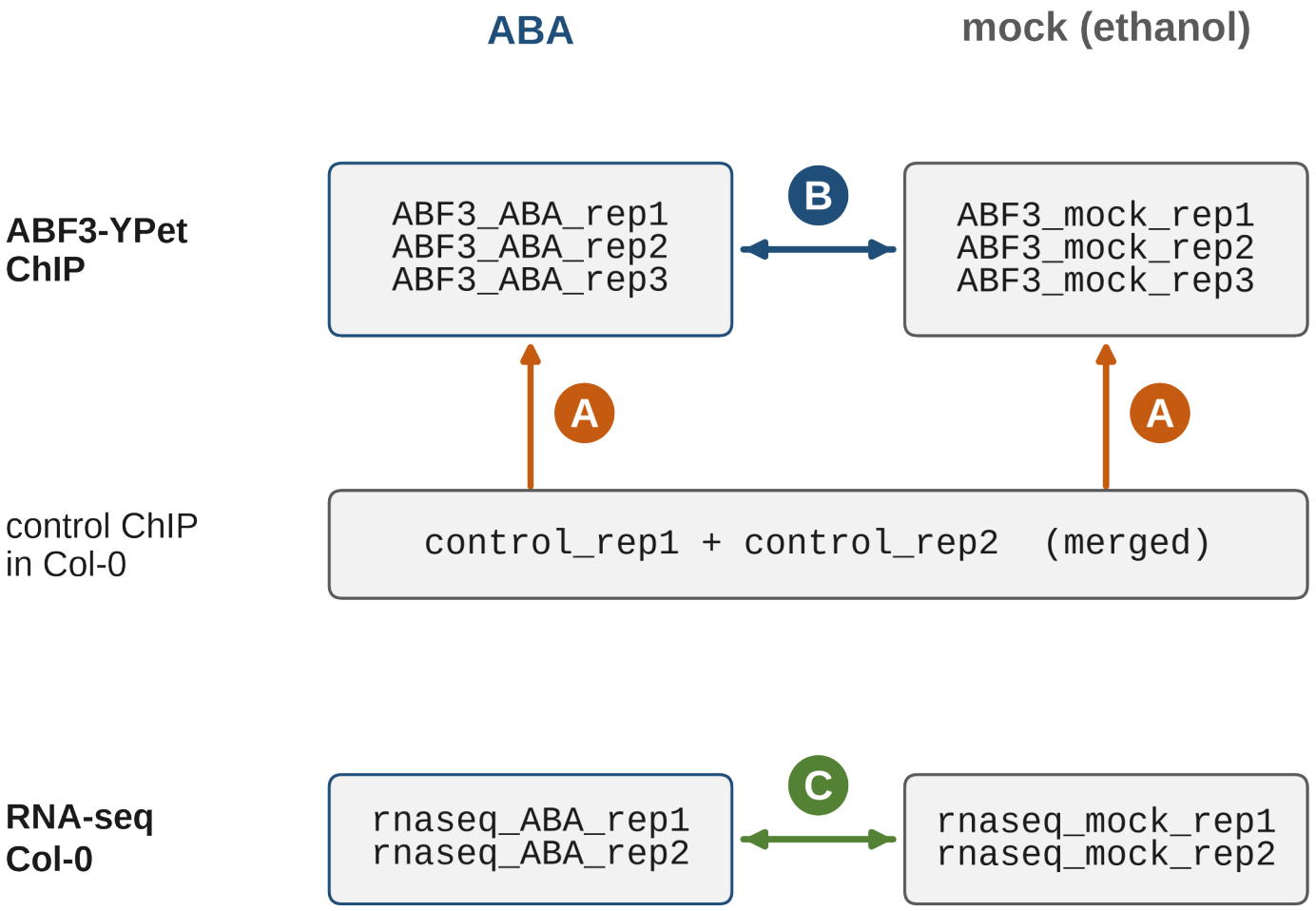
## samples (file names)

| | ABA 4 h | mock 4 h |
|---|---|---|
| **ChIP ABF3-YPet line** | ABF3_ABA_rep1<br>ABF3_ABA_rep2<br>ABF3_ABA_rep3 | ABF3_mock_rep1<br>ABF3_mock_rep2<br>ABF3_mock_rep3 |
| control: ChIP in Col-0 | control_rep1<br>control_rep2 | |
| **RNA-seq Col-0** | rnaseq_ABA_rep1<br>rnaseq_ABA_rep2 | rnaseq_mock_rep1<br>rnaseq_mock_rep2 |

all reads: 50 bp, single-end
ChIP 2.6–3 M reads, RNA-seq 4 M reads per sample

Song et al. 2016, Science 354:aag1550 (GEO GSE80564, GSE80565)

# What is compared with what




**A** **ChIP vs control**

where does ABF3 bind?

peaks: MACS3

**B** **ChIP: ABA vs mock**

does the binding change?

DiffBind

**C** **RNA-seq: ABA vs mock**

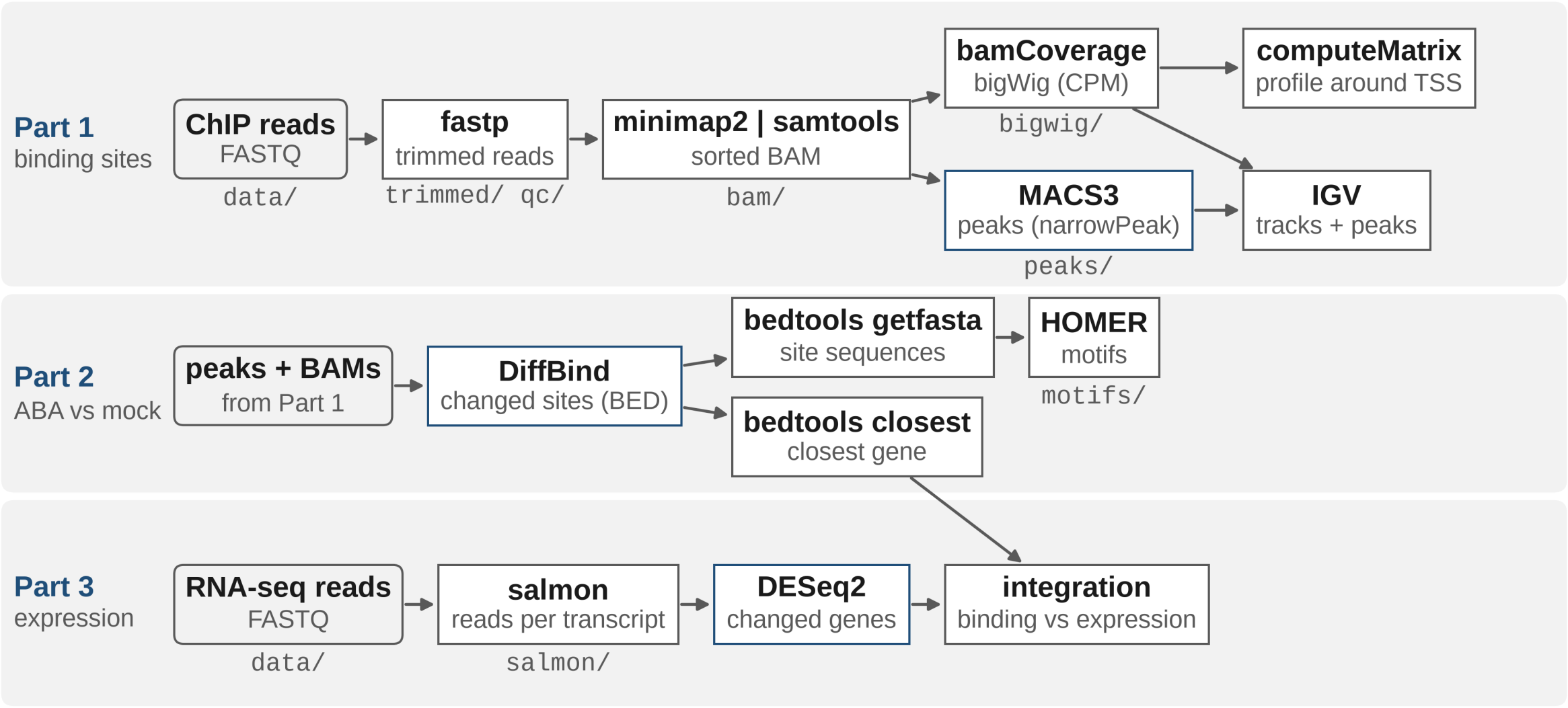
which genes change?

DESeq2

**B + C** are the genes with gained binding induced?

**the control is never compared with ABA or mock directly**

# Three parts, one question



# Part 1 – From reads to binding sites

# fastp: what to read in the report

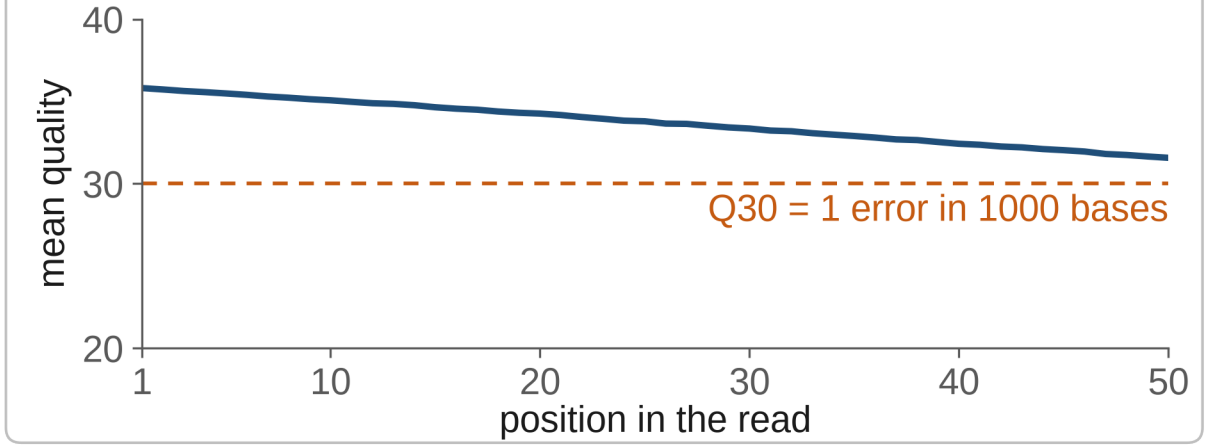
## Summary

| | Before filtering | After filtering |
|---|---|---|
| total reads | 20,000 | 19,989 |
| Q30 bases | 90.2 % | 90.2 % |
| mean length | 50 bp | 50 bp |

**duplication rate:** 8.3 %

reads with low quality: 11 (removed)

## Before filtering: read1: quality




**duplicate = exactly the same sequence as another read**

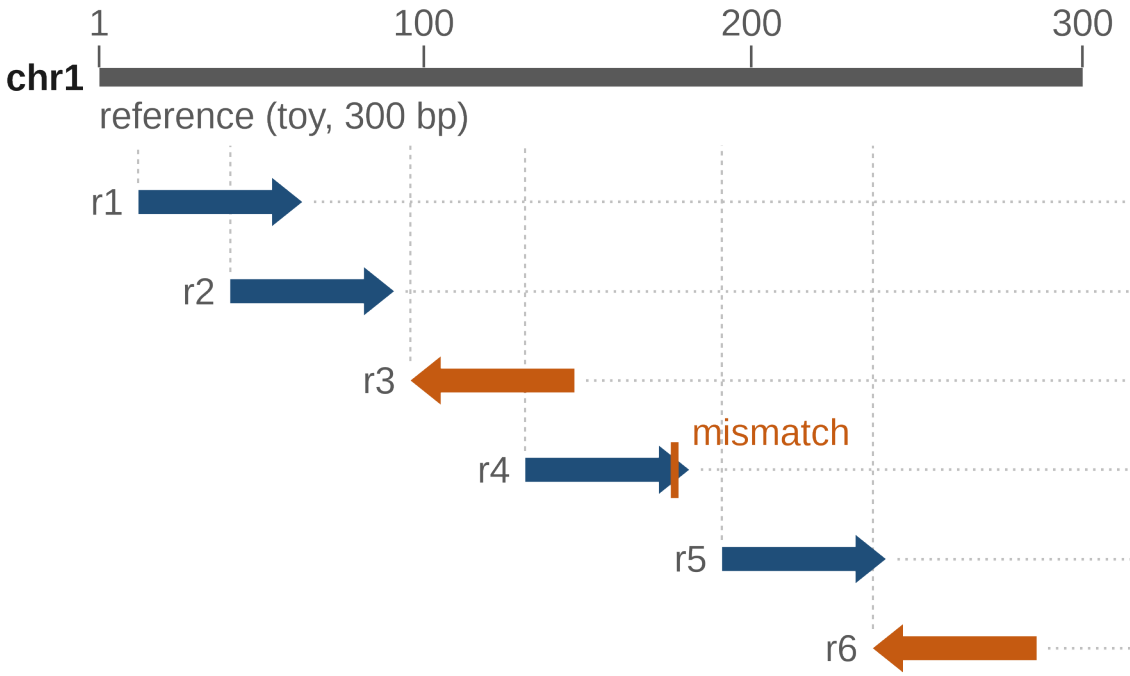
| | | |
|---|---|---|
| read1 | TTTCCTCATGCAATTCAAAACCATGTCCGTAATGTAGGCGAAATAGTAAA | |
| read2 | CCATTTTACGGAGGATACCAAATTCCTCCTTATTCAGGACCTAACCTGAG | |
| read3 | GTAAACCAGGTCTCTCCGCCCCCTTATAAAAGCTGTTGCACCTAGCCAAG | |
| read4 | TTCAACGGCAGCTGCAATGGAAATAGGCAATGACGGATATATATTAAAAA | |
| read5 | GTGTTTTAAGATACATTGAGGCCCGTTCGTGCTCCTCGCCCTGAAGCATT | |
| read6 | GCTTTGTGAAGAGGGACTTCAGCCAATAGACCTGCATACCGGCTCATTCT | |
| read7 | CCATTTTACGGAGGATACCAAATTCCTCCTTATTCAGGACCTAACCTGAG | = read2 |
| read8 | TCTAATAATATACATTTGCTTCGTTGACTAGCAACCCAGGGCTATAGCTA | |
| read9 | TTCCCCCCGCGGCCCACCCAGTATTCCTAACGGAGCATAAATCCCACCCG | |
| read10 | GTGTTTTAAGATACATTGAGGCCCGTTCGTGCTCCTCGCCCTGAAGCATT | = read5 |

**2 duplicates / 10 reads**

**= 20 %**

fastp on these 10 reads:
duplication rate 20.0 %

# Mapping: reads to positions




| QNAME | FLAG | RNAME | POS | MAPQ | CIGAR |
|---|---|---|---|---|---|
| r1 | 0 | chr1 | 13 | 34 | 50M |
| r2 | 0 | chr1 | 41 | 13 | 50M |
| r3 | 16 | chr1 | 96 | 37 | 50M |
| r4 | 0 | chr1 | 131 | 22 | 50M |
| r5 | 0 | chr1 | 191 | 26 | 50M |
| r6 | 16 | chr1 | 237 | 20 | 50M |

FLAG 0 = read on the + strand (→)

FLAG 16 = read on the − strand (←)

POS = first reference base covered by the read

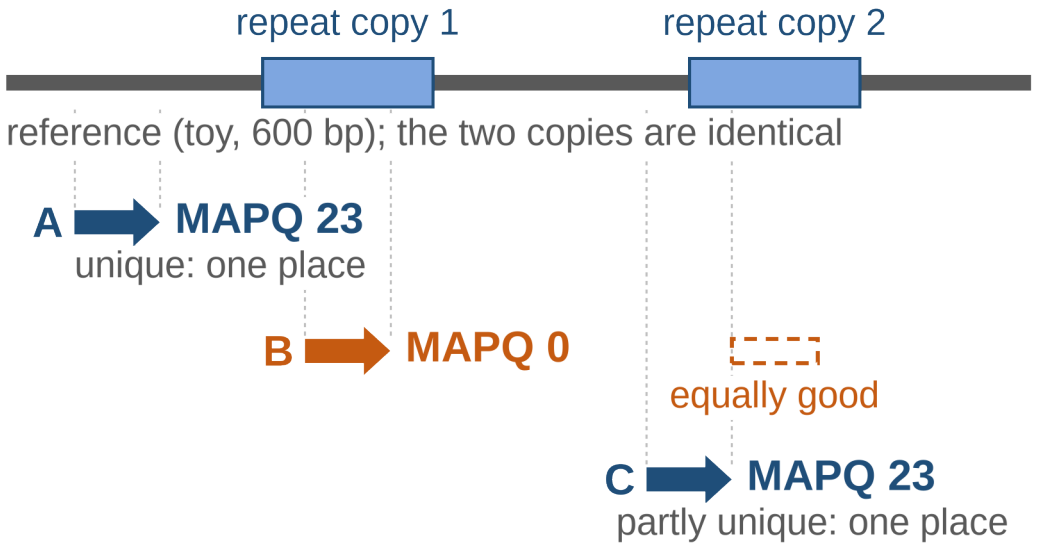
CIGAR 50M = 50 aligned bases

(M = match or mismatch: r4 is 50M too)

MAPQ = how certain the position is

# MAPQ: is the position certain?

- A read from a **repeat** fits several places equally
- The mapper picks one at random and reports **MAPQ ≈ 0**
- Unique read → MAPQ > 0 (minimap2 maximum 60; short reads rarely reach it)
- `--minMappingQuality 10` drops the uncertain ones




| QNAME | FLAG | RNAME | POS | MAPQ | CIGAR |
|---|---|---|---|---|---|
| A | 0 | chr1 | 41 | **23** | 50M |
| B | 0 | chr1 | 176 | **0** | 50M |
| C | 0 | chr1 | 376 | **23** | 50M |

**--minMappingQuality 10 keeps A and C, drops B**

# samtools idxstats: reads per chromosome

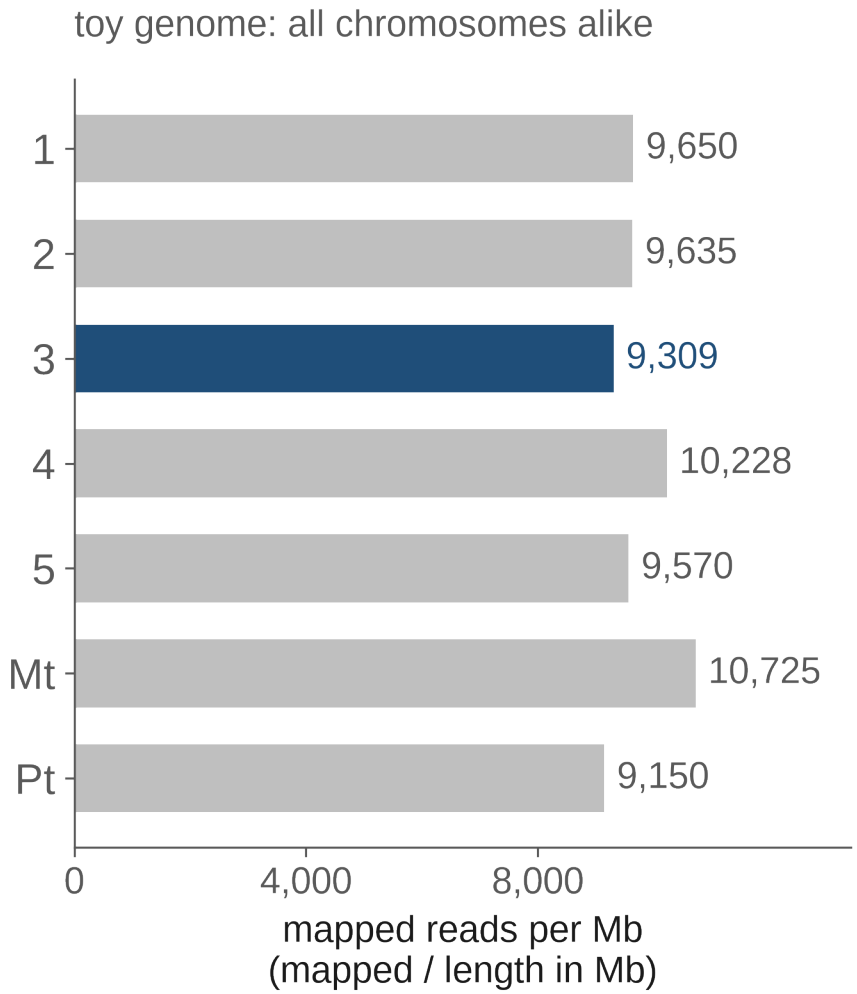
```
$ samtools idxstats sample.bam
```

| name | length | mapped | unmapped | |
|---|---|---|---|---|
| 1 | 300000 | 2895 | 0 | |
| 2 | 200000 | 1927 | 0 | |
| 3 | 230000 | 2141 | 0 | |
| 4 | 180000 | 1841 | 0 | |
| 5 | 270000 | 2584 | 0 | |
| Mt | 40000 | 429 | 0 | |
| Pt | 20000 | 183 | 0 | |
| * | 0 | 0 | 400 | ← reads without a position |

**chromosome 3:**
**share of the mapped reads**

**2,141 / 12,000 = 17.8 %**

12,000 = sum of the mapped column (without the * line)



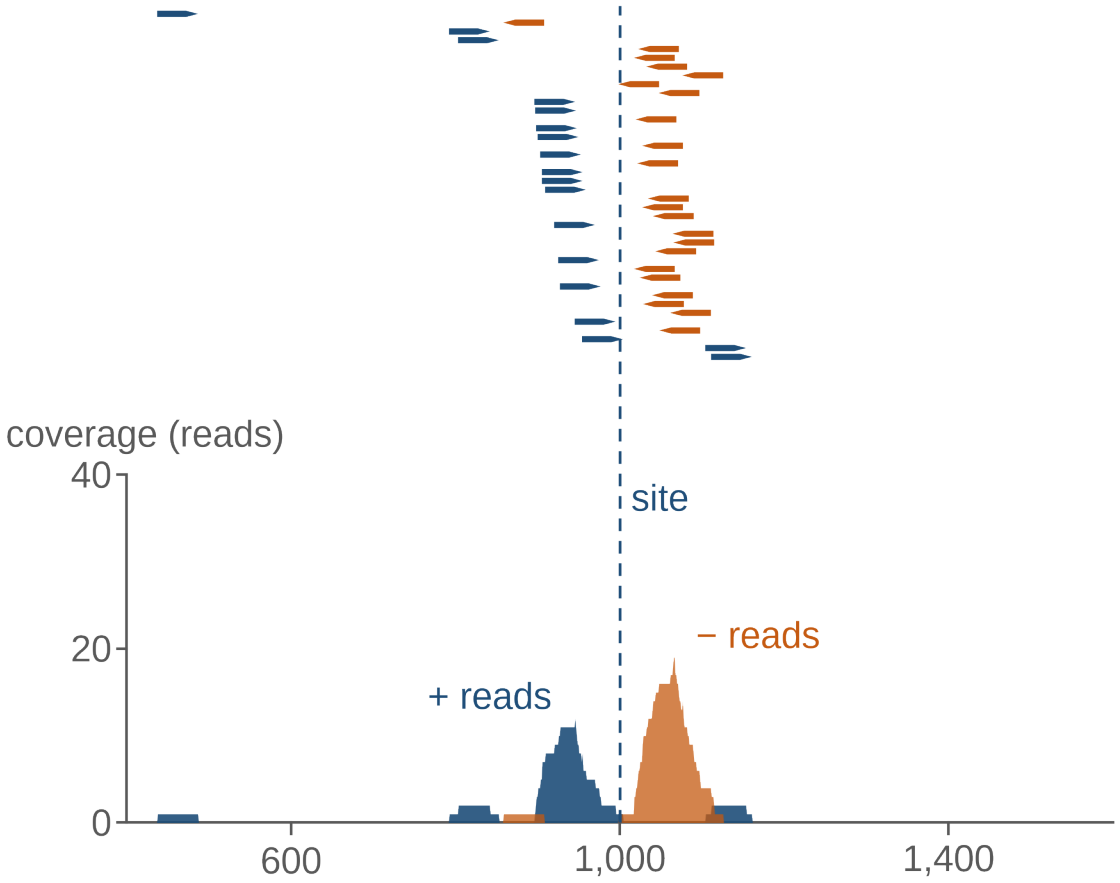

```
$ samtools flagstat sample.bam
12400 + 0 in total (QC-passed reads + QC-failed reads)
12000 + 0 mapped (96.77% : N/A)
```

**12,000 / 12,400 = 96.8 % of all reads mapped**

# Reads are fragment ends

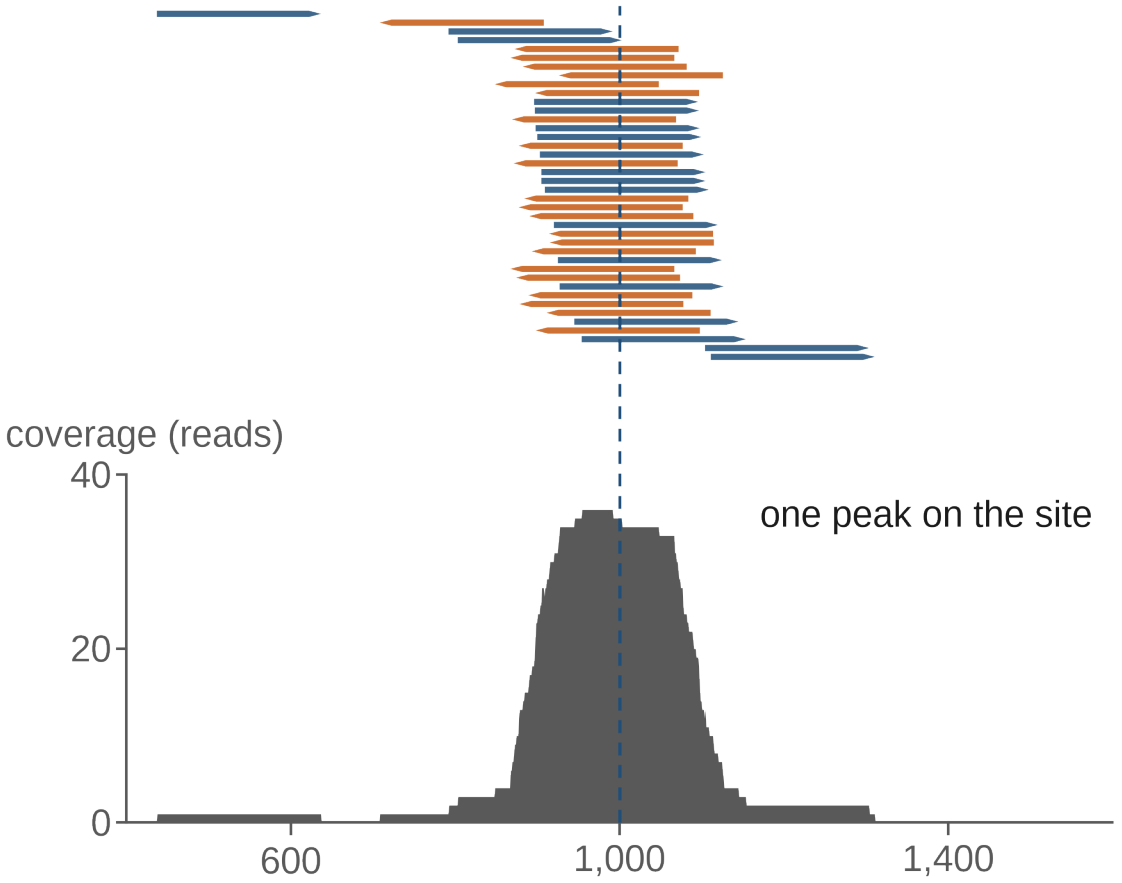
**Reads as mapped (50 bp)**

two humps, either side of the site



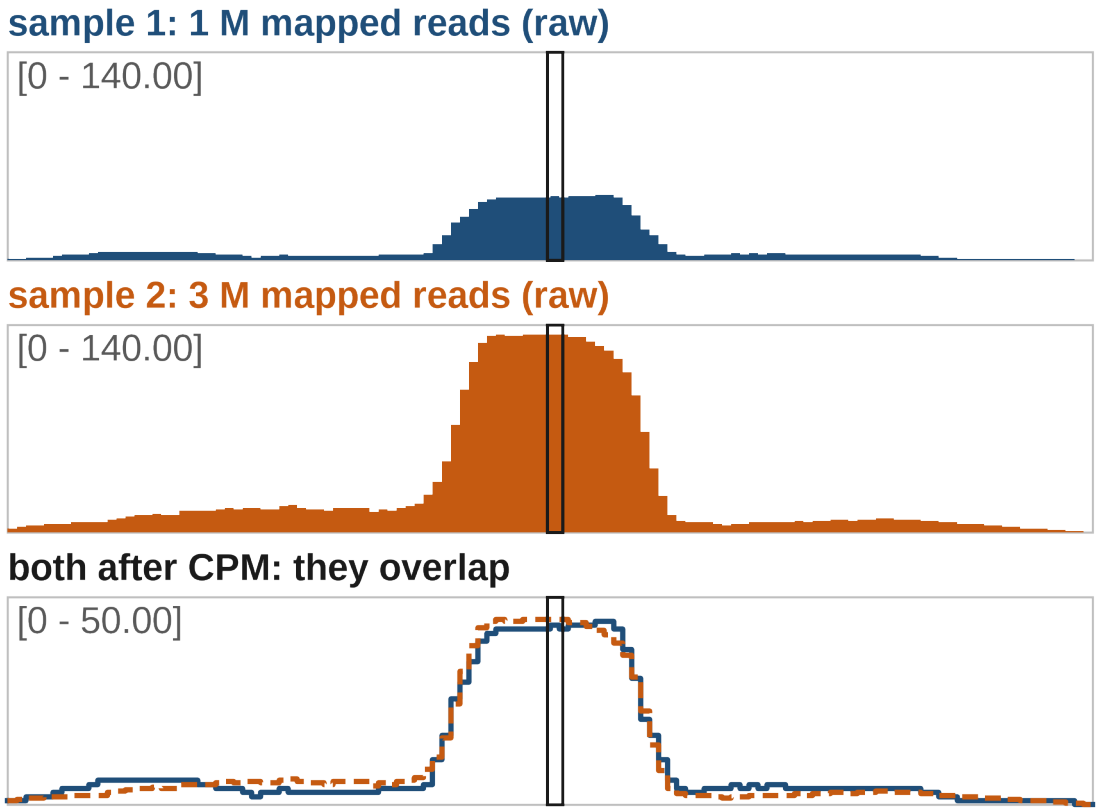

**Each read extended to 200 bp**

`--extendReads 200`



# CPM: comparing samples of different depth

- Sample with 3× more reads → 3× higher coverage everywhere
- **CPM** = reads / million mapped reads
- After CPM, equal binding looks equal
- bigWig = coverage in bins (here 10 bp), small and fast



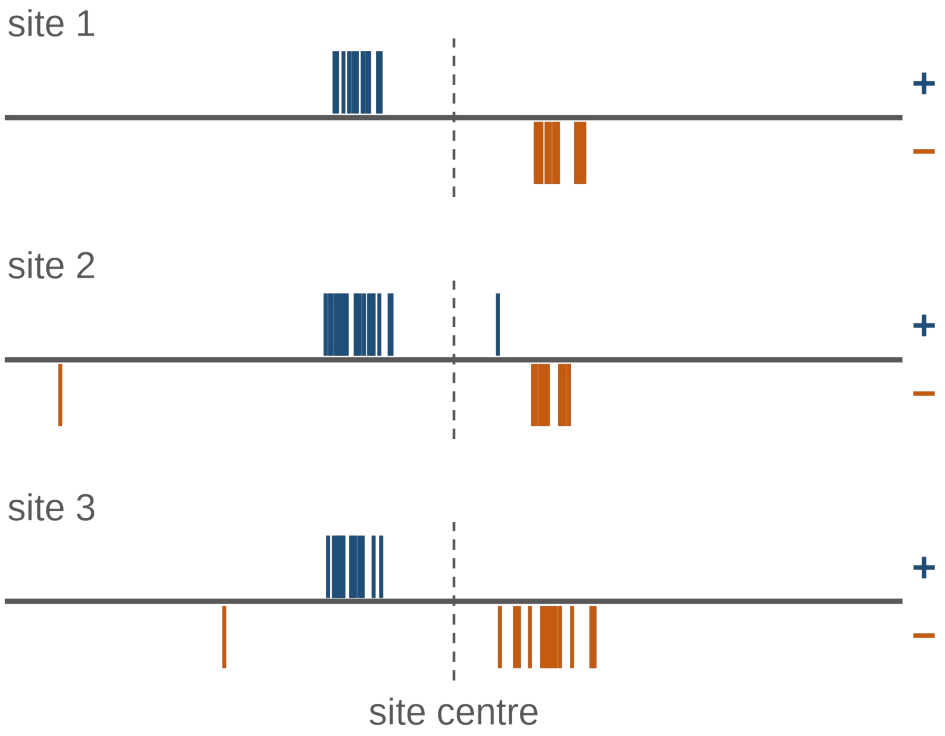

**bin 1,000–1,010:**

sample 1: 44 reads / 1 M = 44.0 CPM

sample 2: 136 reads / 3 M = 45.3 CPM

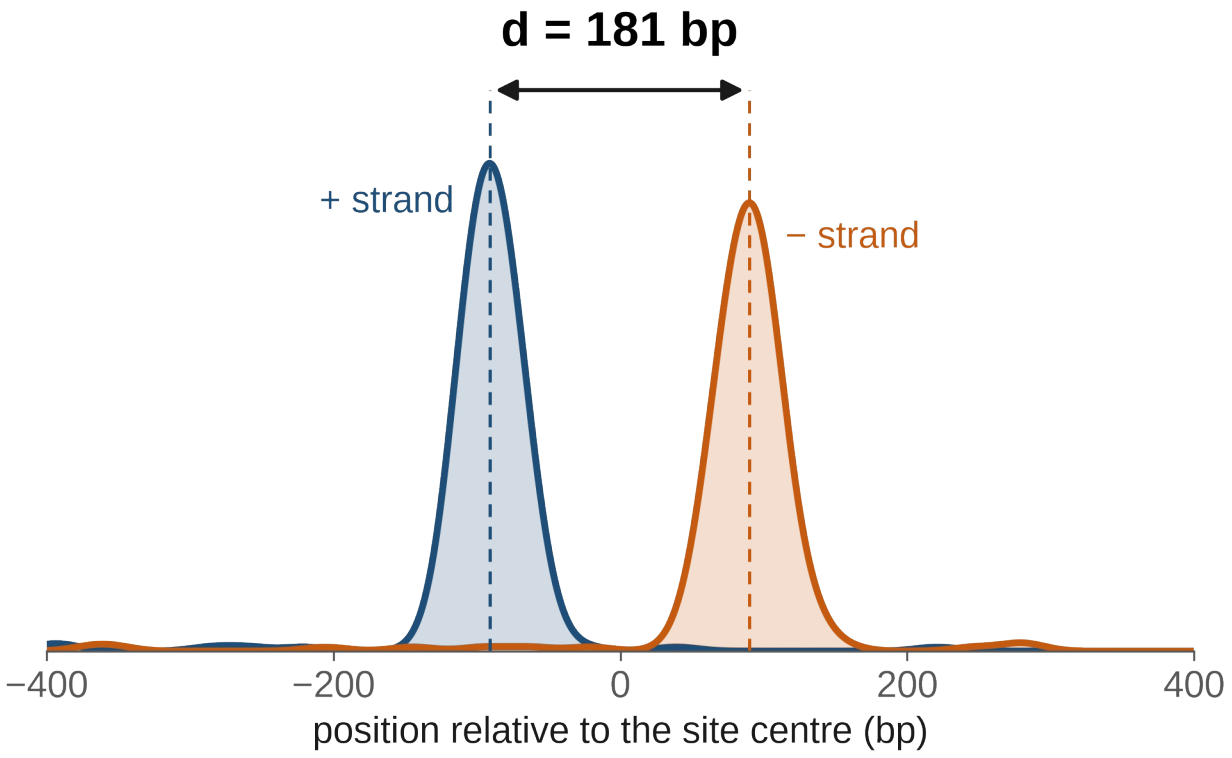
# MACS3: estimating the fragment length

**Read 5' ends at single strong sites**




… and so on for many sites

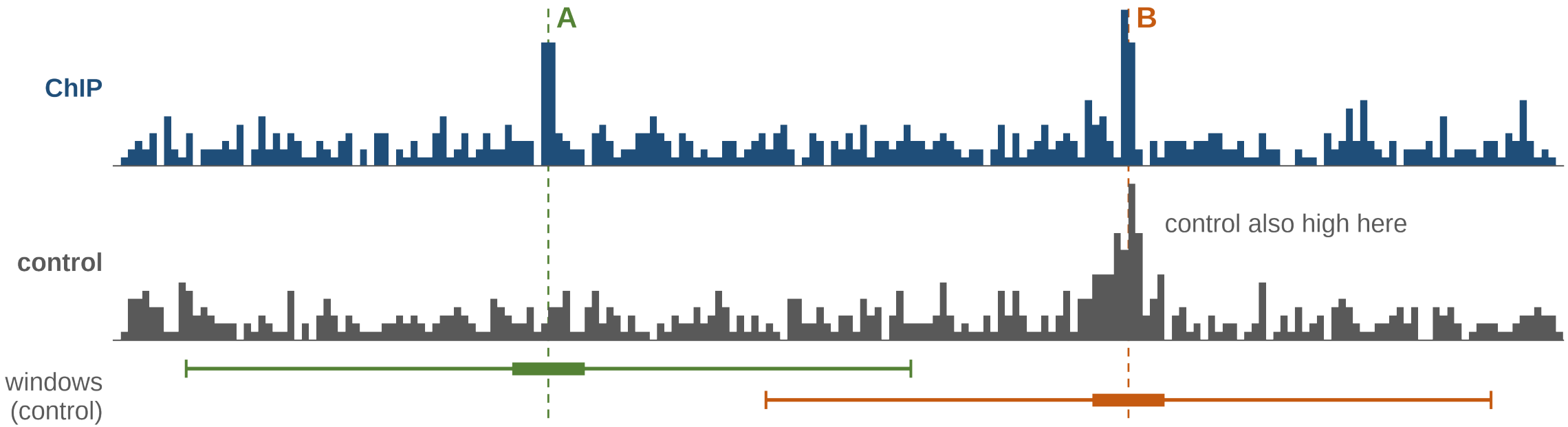
**Averaged over 12 sites**




```
NAME_peaks.xls:   # d = 181
```

Zhang et al. 2008, Genome Biol 9:R137

# MACS3: more reads than expected?




| | ChIP reads | λ genome | λ 1 kb | λ 10 kb | λ local | p | q | peak? |
|---|---|---|---|---|---|---|---|---|
| A | 30 | 5.0 | 5.4 | 5.0 | 5.4 | 2e-13 | 2e-11 | yes |
| B | 34 | 5.0 | 19.2 | 6.3 | 19.2 | 0.0014 | 0.071 | no |

per 200 bp window

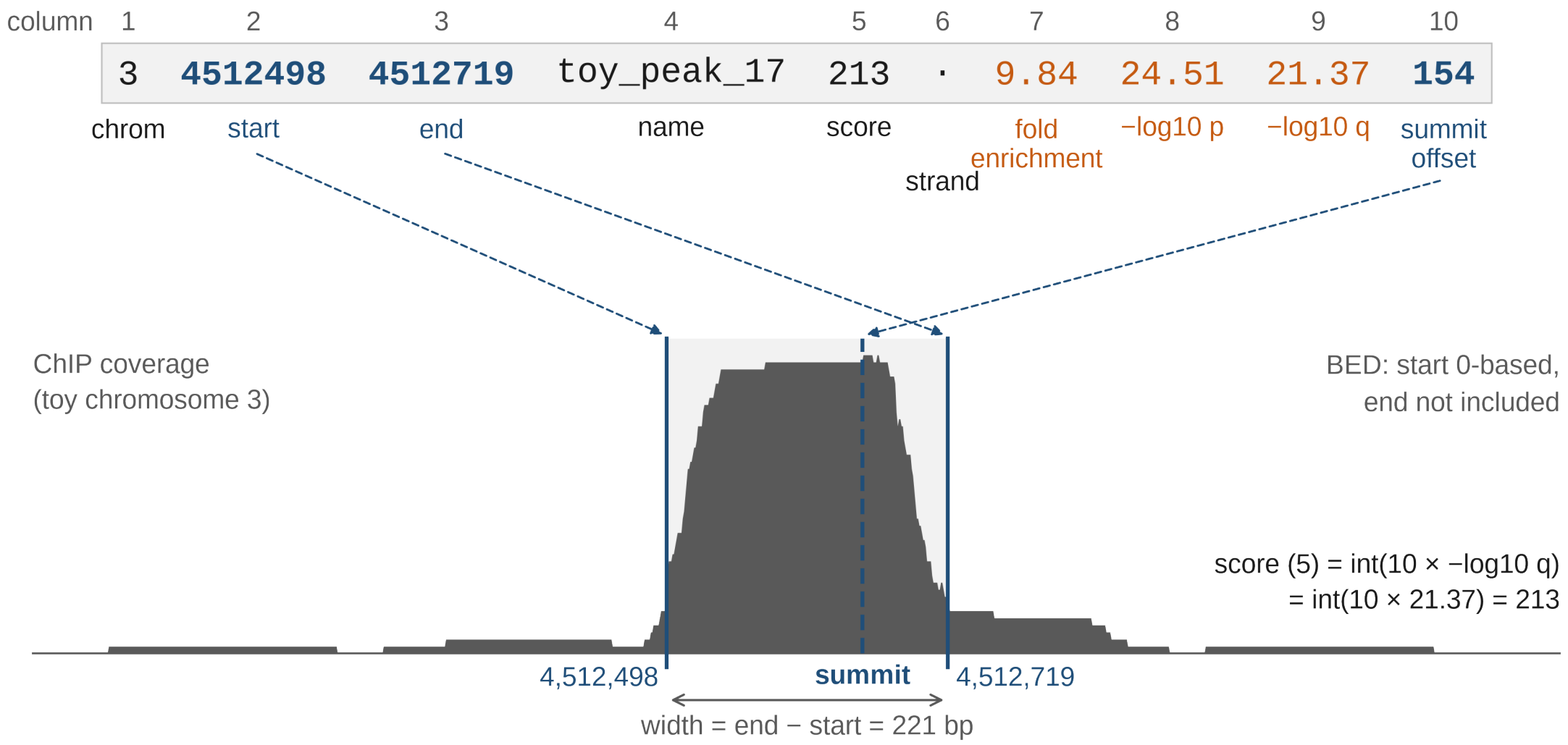
**λ local = the largest**

peak if q < 0.01

λ genome = 3 M reads × 200 / 1.2e8 (-g) = 5.0 p = Poisson probability of ≥ that many reads q = p corrected for many windows

Zhang et al. 2008, Genome Biol 9:R137; MACS3 callpeak documentation

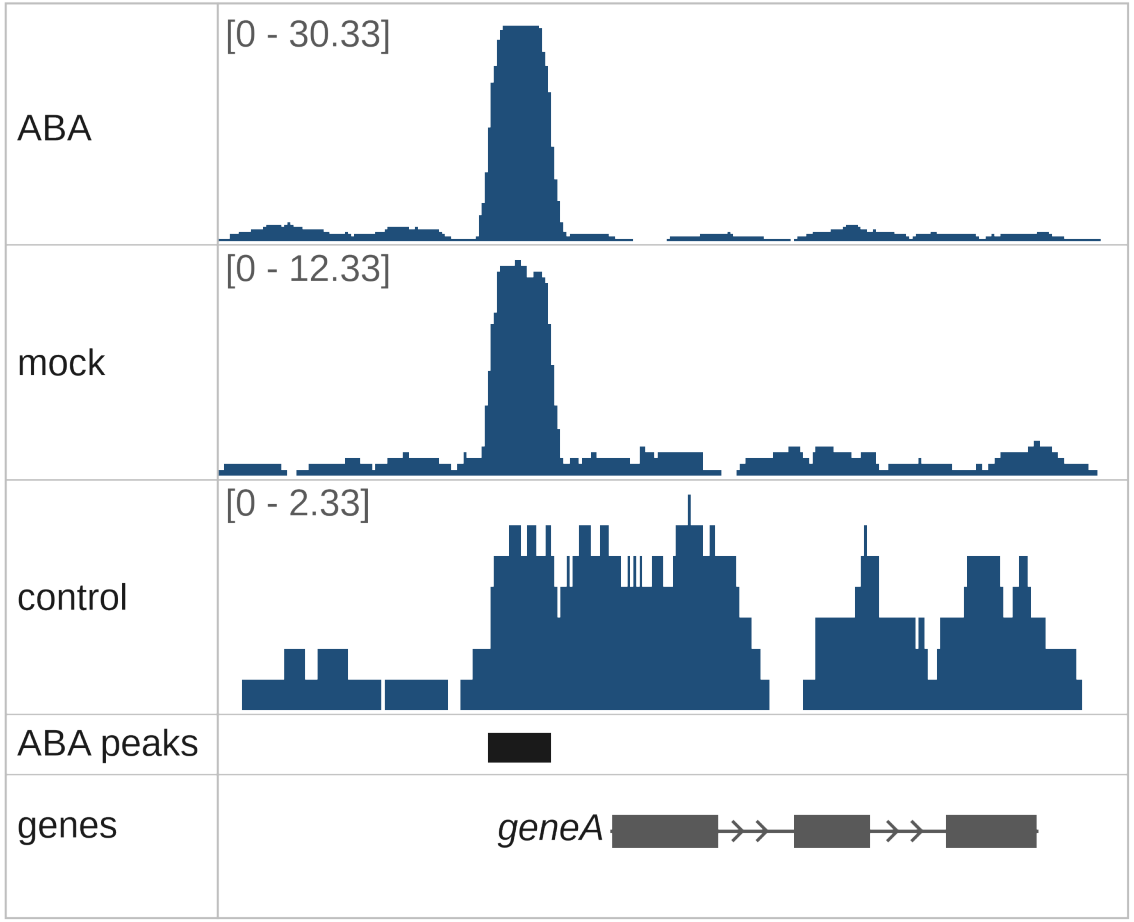
# A narrowPeak line



# IGV: compare tracks on one scale

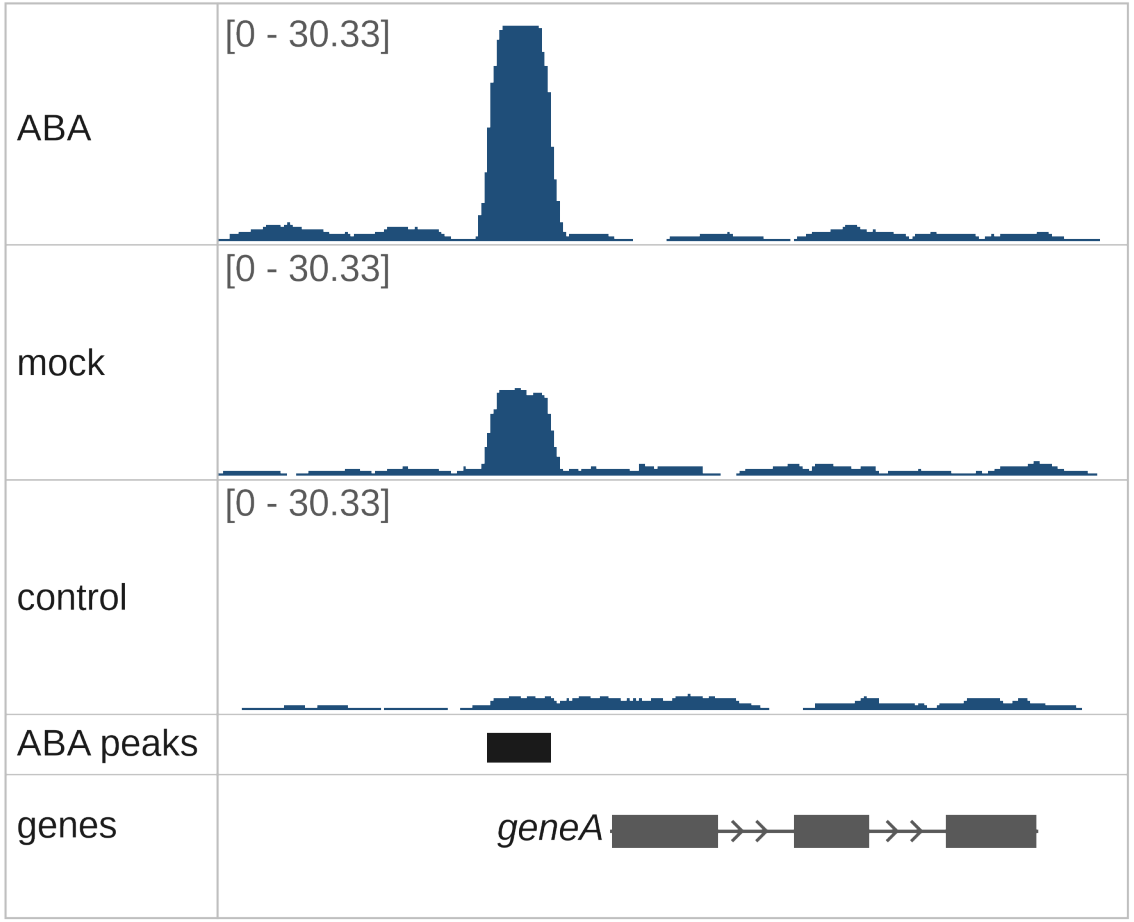
**Autoscale (each track alone)**

IGV default: every track fills its height

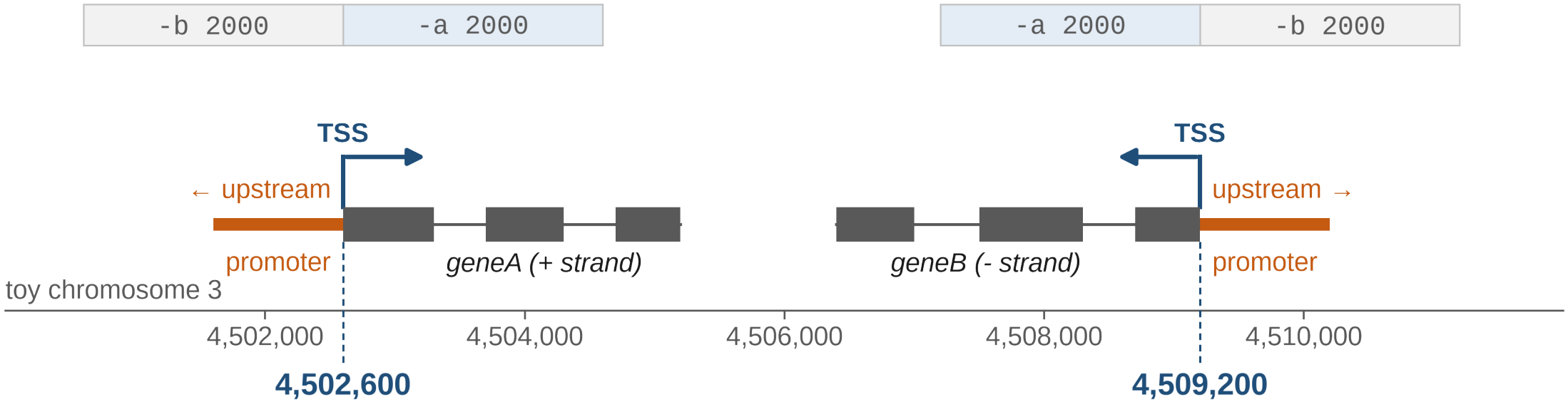



**Group Autoscale (one range)**

select tracks → right-click → Group Autoscale



# Genes have a direction

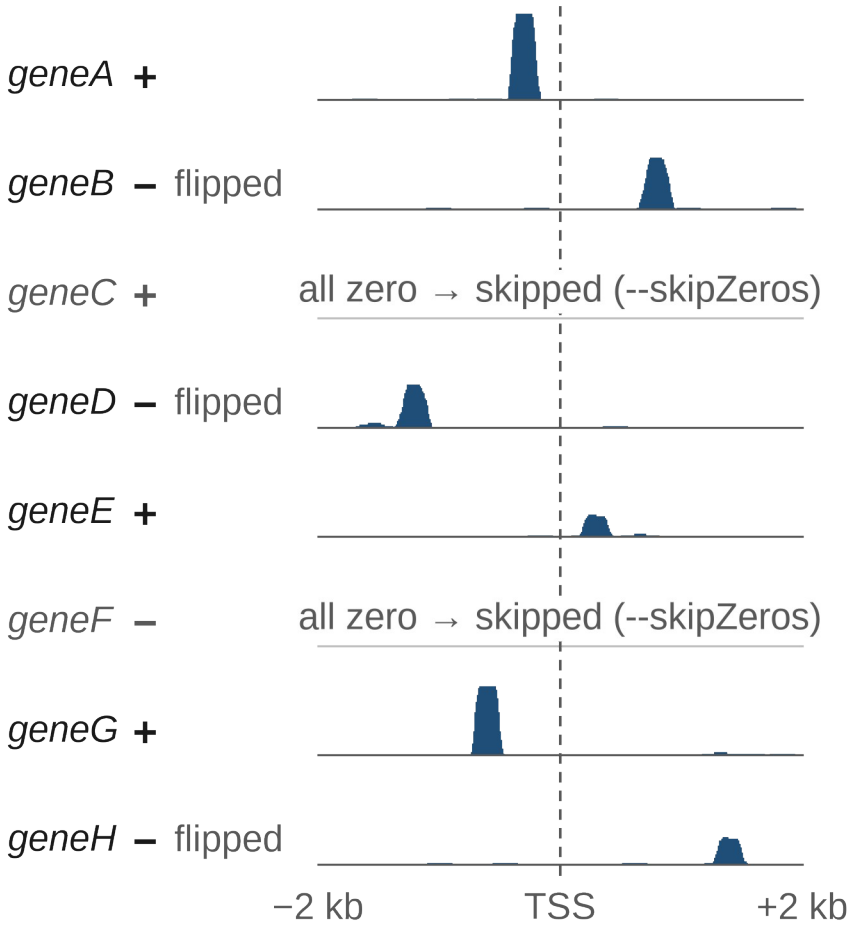



TAIR10_genes.bed-style lines

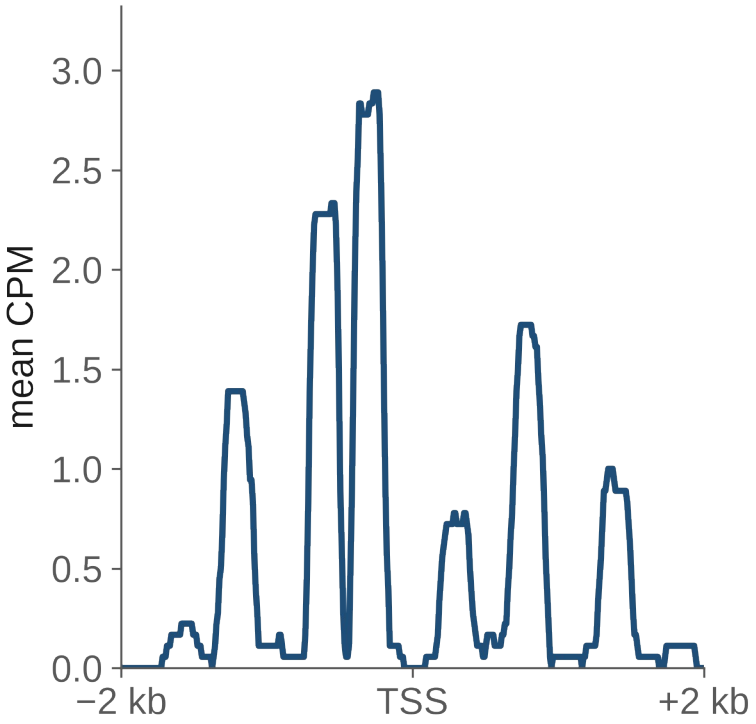
| | | | | | | |
|---|---|---|---|---|---|---|
| 3 | **4502600** | 4505200 | AT3G99010 | geneA | + | TSS = start |
| 3 | 4506400 | **4509200** | AT3G99020 | geneB | - | TSS = end |

# computeMatrix: signal around many genes

**1 window around each TSS**
(− strand genes flipped)

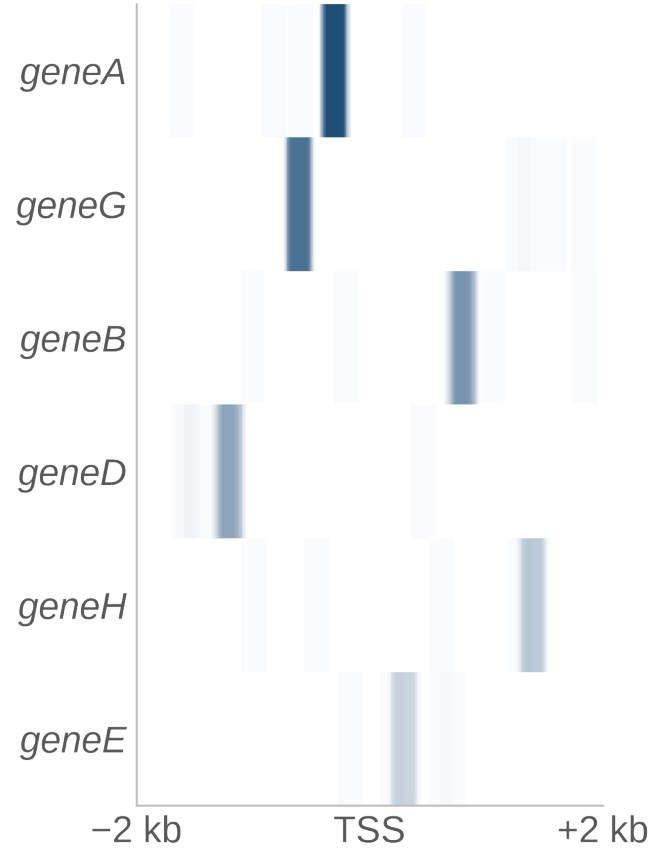



**2 average = profile**
`plotProfile`




mean of the 6 kept rows
(one sample, toy genes)

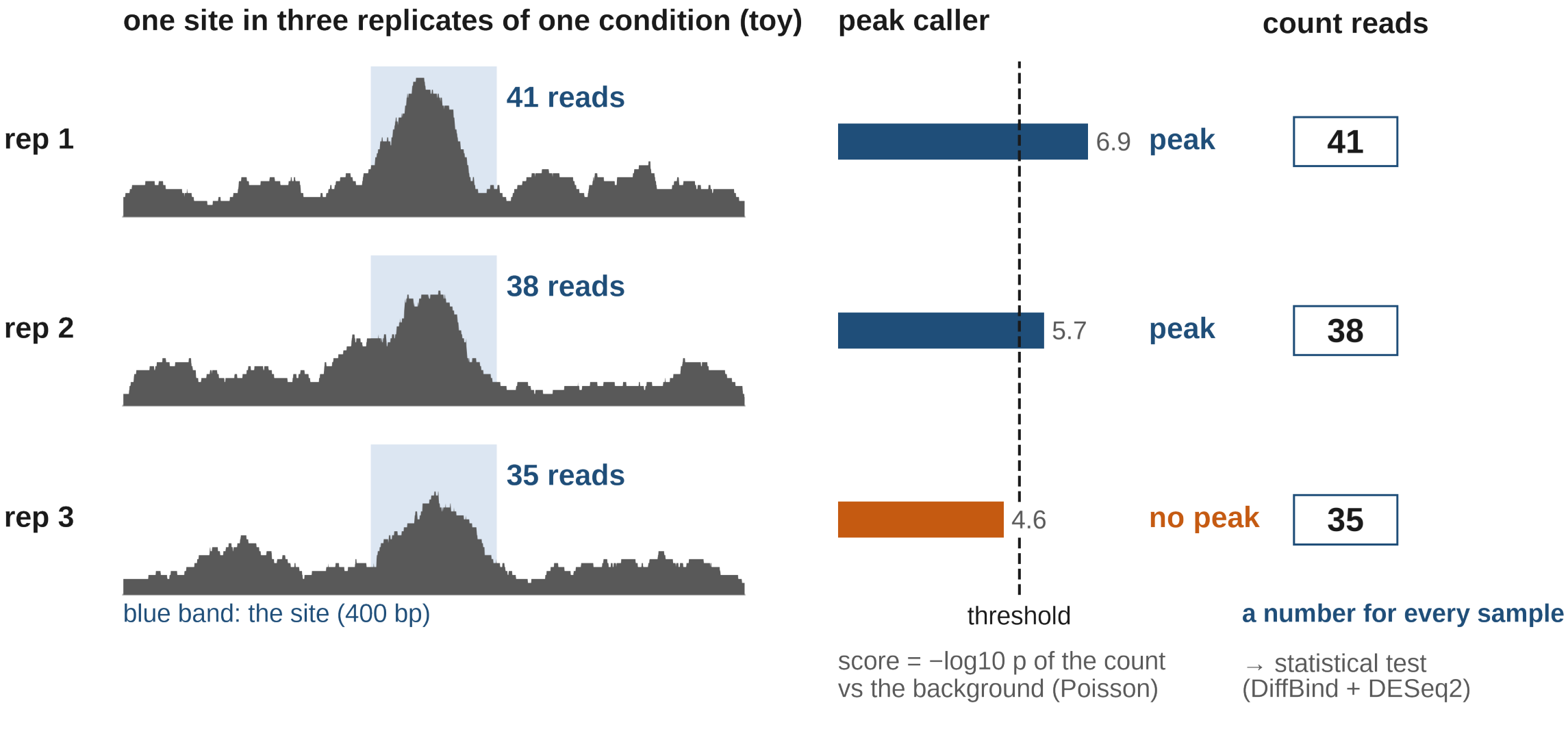
**3 rows as colour**
`plotHeatmap`




sorted: strongest on top

# Part 2 – Differential binding

# Why not just count peaks?



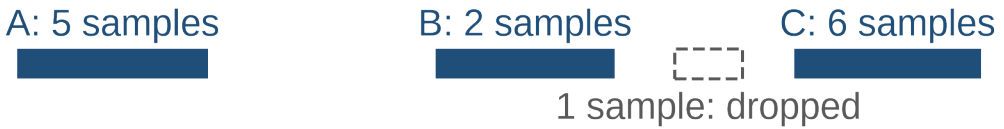
# DiffBind: from peaks to a test

**1 peaks of each sample**




bar = peak
white line = summit

**2 consensus**




merged; kept if in
≥ 2 samples (minOverlap)

**3 summit ± 200 bp**




summits = 200:
all sites the same width

**4 count the reads of every site in every sample**

| site | ABA 1 | ABA 2 | ABA 3 | mock 1 | mock 2 | mock 3 |
|---|---|---|---|---|---|---|
| A | 88 | 114 | 96 | 36 | 41 | 30 |
| B | 35 | 35 | 30 | 37 | 33 | 47 |
| C | 29 | 43 | 40 | 83 | 75 | 120 |

→ normalise by library size → DESeq2 →

| log2 FC | FDR |
|---|---|
| +1.48 | 2.4e-06 |
| -0.22 | 1.00 |
| -1.31 | 2.2e-04 |

one test per site

# The sample sheet

```
SampleID,Condition,Replicate,bamReads,ControlID,bamControl,Peaks,PeakCaller
ABA_rep1,ABA,1,bam/ABF3_ABA_rep1.bam,control,bam/control.bam,peaks/ABF3_ABA_rep1_peaks.narrowPeak,narrow
…  ABA_rep2, ABA_rep3
mock_rep1,mock,1,bam/ABF3_mock_rep1.bam,control,bam/control.bam,peaks/ABF3_mock_rep1_peaks.narrowPeak,narrow
```

PeakCaller narrow:
MACS narrowPeak format

**SampleID, Condition, Replicate**

| | | | |
|---|---|---|---|
| **ABA** | 1 | 2 | 3 |
| **mock** | **1** | 2 | 3 |

the groups to compare (ABA vs mock) and the replicates in each

**bamReads**

`bam/ABF3_ABA_rep1.bam`

…

**`bam/ABF3_mock_rep1.bam`**

…

mapped reads of this ChIP sample (minimap2 + samtools sort)

**ControlID, bamControl**

**`bam/control.bam`**

= control_rep1 + rep2 (samtools merge)

the same control in all six lines

**Peaks**

**`peaks/ABF3_mock_rep1_peaks.narrowPeak`**

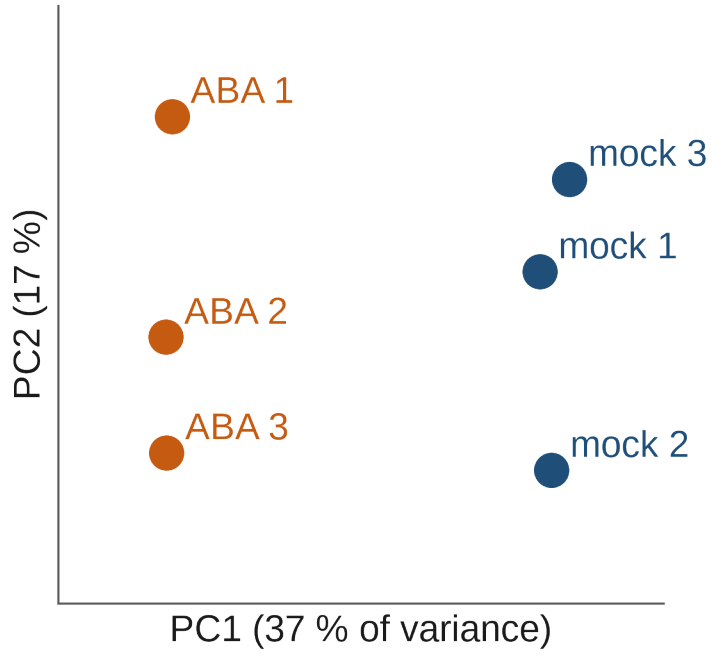


MACS3 peaks of this sample: DiffBind builds the consensus sites from them

6 lines = 6 ChIP samples (3 ABA + 3 mock); the control is not a line of its own

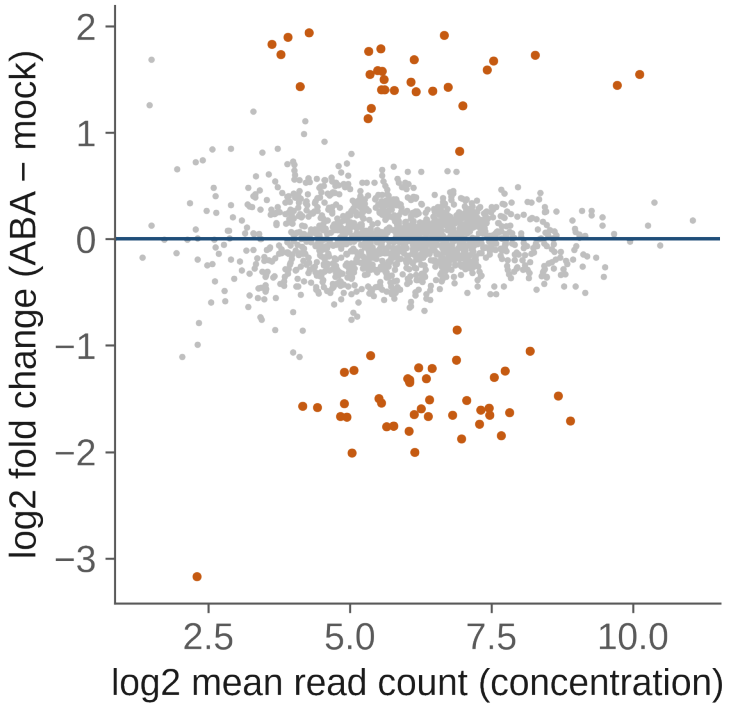
# Reading DiffBind plots

**PCA (dba.plotPCA)**



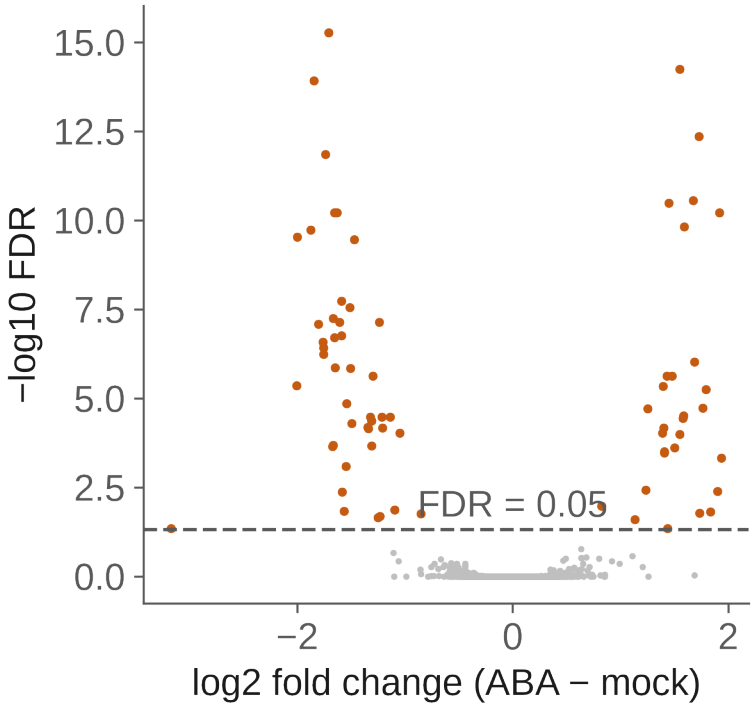

one point = one sample
close = similar read counts over all sites

**MA plot (dba.plotMA)**




one point = one site
orange: FDR < 0.05
above 0: more reads in ABA

**volcano (dba.plotVolcano)**

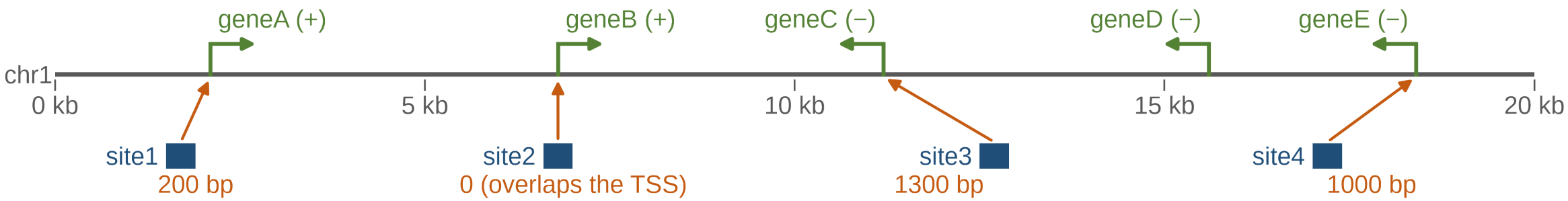



one point = one site
right: stronger in ABA
higher: more significant

# bedtools closest: the nearest gene

```
bedtools closest -a sites.bed -b tss.bed -d -t first
```

-d  add the distance in bp as the last column  -t first  if two TSS are equally close, report only the first




**output (one line per site):**

```
chr1  1500   1901   site1  1.85  0.0004  chr1  2100   2101   TG001  geneA  +  200
chr1  6600   7001   site2  0.42  0.31    chr1  6800   6801   TG002  geneB  +  0
chr1  12500  12901  site3  2.1   0.0001  chr1  11200  11201  TG003  geneC  -  1300
chr1  17000  17401  site4  -0.9  0.04    chr1  18400  18401  TG005  geneE  -  1000
```

**A: the site (columns 1–6)**  **B: the closest TSS (columns 7–12)**  **distance (bp)**

# Motifs: what is common to the sites?

### 12 target sequences (sites)

1 CAGTAATA**TTGACT**AGGAAATGATAATTTA
2 TT**AGTCAA**CGATATCTTAATTTACATTATA (− strand)
3 TAGTTGCGTGCCAATAATCTGCTGCTAATG
4 ACAGGATATT**TTGACT**TGAACGTTAATTCT
5 TTGCTGTC**TTGACC**GAACATAGGGAAGGTC
6 AGTCTGCTGAAAAAATAAGTACCAAGTCTT
7 CATTTTTTAGT**AGTCAA**GTGAAAAGAACTA (− strand)
8 AATATACTAAATGATCCT**ATGACT**GGCAAA
9 TCTCGTATA**TTGACT**ATCTATCTCACATTT
10 CTATAAGTAATGCTAACGCATTCGTCCAAA
11 AGACAGTT**AGTCTA**TAGTAG**GGTCAA**GGGT (− strand)
12 ATAAAAATGTAT**TTGACT**TAAATTATTAGA

**+ 60 background sequences**

(random regions of the same size)

### count each 6-mer (both strands)

| 6-mer | targets | | background | | p |
|---|---|---|---|---|---|
| TTGACT | 6/12 | 50 % | 1/60 | 2 % | 4e-5 |
| TGACTA | 4/12 | 33 % | 0/60 | 0 % | 5e-4 |
| CTTGAC | 3/12 | 25 % | 1/60 | 2 % | 1e-2 |
| AATGAT | 2/12 | 17 % | 0/60 | 0 % | 3e-2 |
| ACTAAA | 2/12 | 17 % | 0/60 | 0 % | 3e-2 |
| AGACAG | 2/12 | 17 % | 0/60 | 0 % | 3e-2 |

… ranked by enrichment p-value

(hypergeometric test)

### words ≤ 1 mismatch from TTGACT

TTGACT
TTGACT
TTGACT
TTGACC
TTGACT
TTCACT
ATGACT
TTGACT
TAGACT
TTGACC
TTGACT

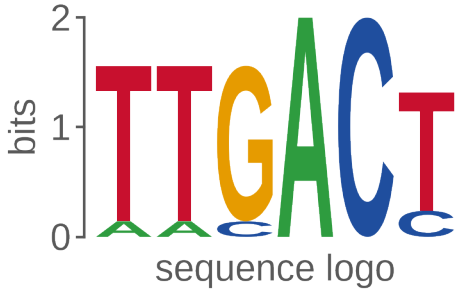
→

| | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| A | 1 | 1 | 0 | 11 | 0 | 0 |
| C | 0 | 0 | 1 | 0 | 11 | 2 |
| G | 0 | 0 | 10 | 0 | 0 | 0 |
| T | 10 | 10 | 0 | 0 | 0 | 9 |

counts (of 11)

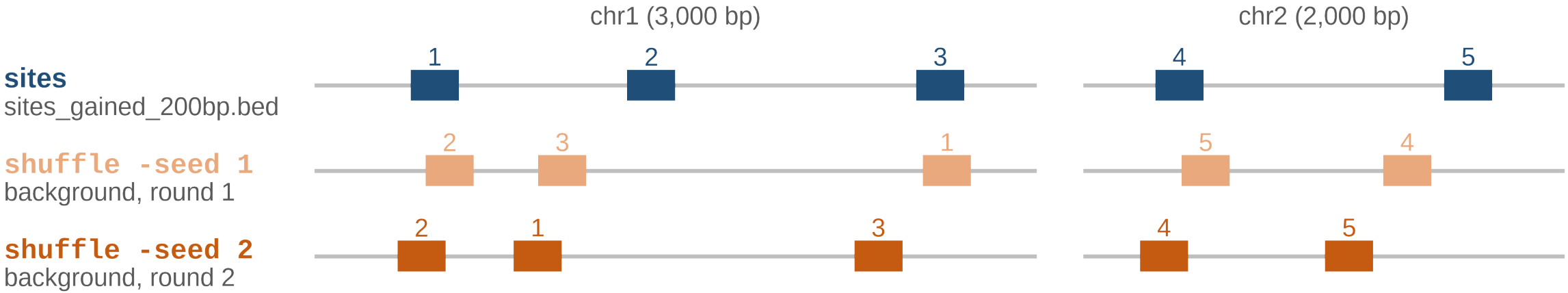
**TTGACT: in 50 % of targets, 2 % of background → motif candidate**

HOMER then lets in similar words (mismatches) while the enrichment improves → a motif matrix




sequence logo

# Background: random regions like the sites



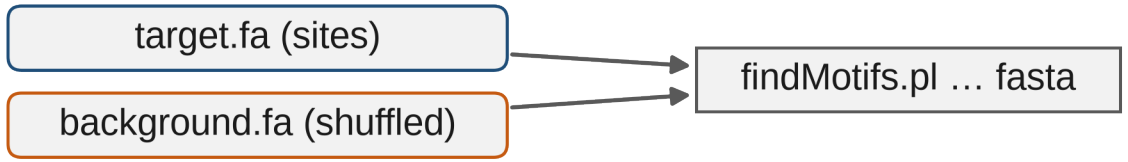

same number, same length (200 bp), same chromosome (-chrom), not overlapping each other (-noOverlapping)

```
bedtools getfasta -fi genome.fa -bed regions.bed
```

```
chr1   400   600   site1   1.0   .
```
one BED line (0-based start, end not included)

→

**>chr1:400-600**
ATTTTCATAAGCATAGTTAAACCTTTTTAGTAAGTGAGGATAGTTTTCGCTAAT…
one FASTA record: name = chrom:start-end, 200 bp on one line




in the exercise: seeds 1 … 5 → 5 × as many background regions

# Reading HOMER results

homerResults.html  Total target sequences = 12  Total background sequences = 56

| Rank | Motif | P-value | log P-pvalue | % of Targets | % of Background | STD (Bg STD) | Best Match/Details |
|---|---|---|---|---|---|---|---|
| 1 * |  | 1e-5 | -1.296e+01 | **66.67%** | **4.91%** | 4.7bp (5.7bp) | ToyTF/made-up-example/Homer(0.987)<br>**similarity score 0.987**<br>More Information |
| 2 * |  | 1e-2 | -5.960e+00 | **33.33%** | **3.21%** | 9.6bp (3.5bp) | ToyTF/made-up-example/Homer(0.345)<br>**similarity score 0.345**<br>More Information |

sorted by p-value

the motif found de novo (logo)

enrichment in targets vs background

natural log of p

targets with the motif

background sequences with motif

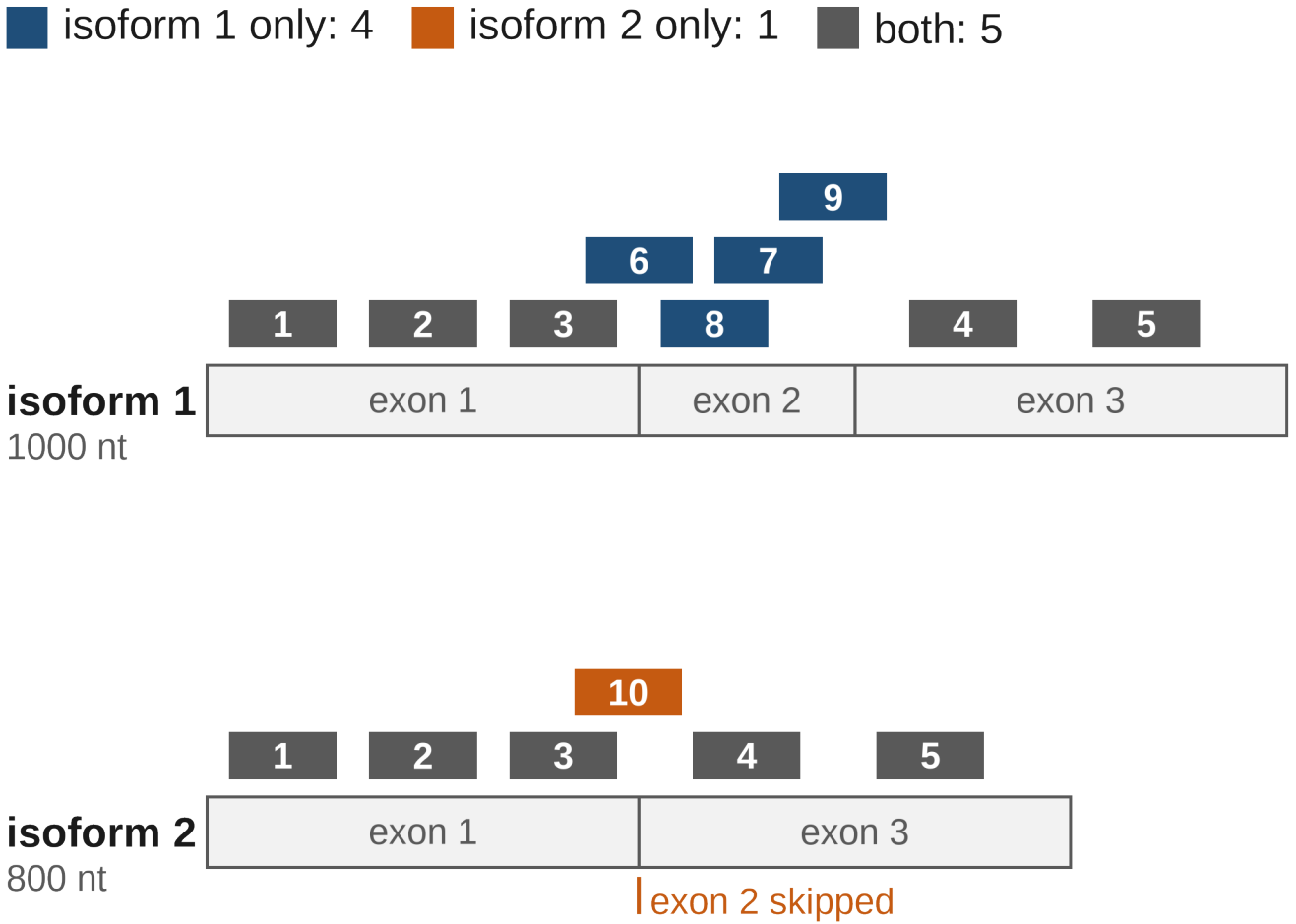
most similar KNOWN motif (here a made-up library with one motif, ToyTF)

score 0–1, 1 = identical: a suggestion, not an identification — compare the logos (More Information)

* = HOMER's flag "possible false positive"

# Part 3 – Binding and expression

# salmon: reads to transcripts




**estimated reads (EM)**

| round | isoform 1 | isoform 2 |
|---|---|---|
| start | 6.50 | 3.50 |
| 1 | 6.99 | 3.01 |
| 2 | 7.25 | 2.75 |
| 3 | 7.39 | 2.61 |
| … | | |
| 35 (stable) | 7.57 | 2.43 |

start: shared reads split half and half
each round: split by reads per nt
of each isoform (isoform 1 is longer)

**gene = 7.57 + 2.43 = 10 reads**

Patro et al. 2017, Nat Methods 14:417

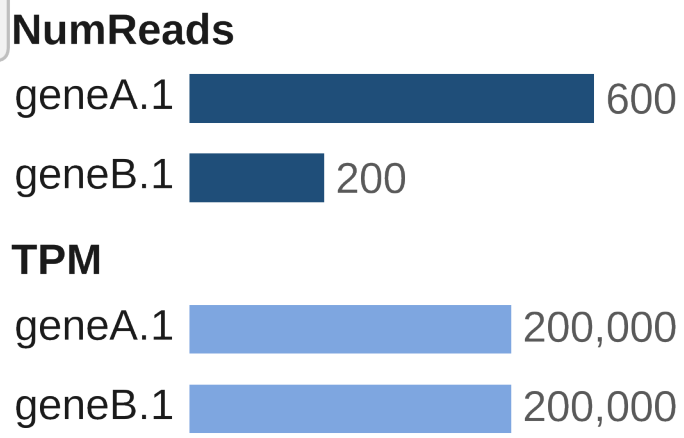
# quant.sf and TPM

transcript | length (nt) | corrected for fragment length | transcripts per million | estimated reads

| Name | Length | EffectiveLength | TPM | NumReads |
|---|---|---|---|---|
| geneA.1 | 3250 | 3000.000 | 200000.000000 | 600.000 |
| geneB.1 | 1250 | 1000.000 | 200000.000000 | 200.000 |
| geneC.1 | 750 | 500.000 | 600000.000000 | 300.000 |

quant.sf

| Name | NumReads ÷ EffectiveLength | rate | rate ÷ S × $10^6$ = TPM |
|---|---|---|---|
| geneA.1 | 600 ÷ 3000 | 0.2 | 200,000 |
| geneB.1 | 200 ÷ 1000 | 0.2 | 200,000 |
| geneC.1 | 300 ÷ 500 | 0.6 | 600,000 |
| | | **S = 1.0** | **sum = 1,000,000** |




**geneA.1: 3× the effective length**
**3× more reads, same TPM**

# DESeq2: which genes changed?

**counts per gene**

| | ABA | | mock | |
|---|---|---|---|---|
| **gene** | **1** | **2** | **1** | **2** |
| geneA | 652 | 1,187 | 205 | 138 |
| geneB | 180 | 2,300 | 330 | 60 |
| geneC | 790 | 1,520 | 1,010 | 690 |
| geneD | 41 | 77 | 48 | 36 |
| geneE | 2,410 | 4,480 | 3,050 | 2,080 |
| all reads | 461k | 871k | 587k | 394k |
| **size factor** | **0.84** | **1.56** | **1.06** | **0.73** |

median of ratios to the average sample, over all genes (toy: 505 genes)

÷ →

**normalised counts**

| | ABA | | mock | |
|---|---|---|---|---|
| | **1** | **2** | **1** | **2** |
| geneA | 778 | 763 | 194 | 189 |
| geneB | 215 | 1,478 | 313 | 82 |
| geneC | 943 | 977 | 956 | 945 |
| geneD | 49 | 49 | 45 | 49 |
| geneE | 2,877 | 2,880 | 2,888 | 2,848 |

geneA: replicates agree

geneB: replicates disagree

**same fold change, different padj**

→

**DESeq2 output, ABA vs mock**

| | **baseMean** | **log2FoldChange** | **pvalue** | **padj** |
|---|---|---|---|---|
| geneA | 481 | 2.01 | 2e-18 | 8e-16 |
| geneB | 522 | 2.10 | 0.14 | 0.99 |
| geneC | 955 | 0.01 | 0.94 | 0.99 |
| geneD | 48 | 0.06 | 0.90 | 0.99 |
| geneE | 2,873 | 0.01 | 0.98 | 0.99 |

baseMean: mean of the normalised counts

log2FoldChange ≈ log2(ABA ÷ mock)

padj: pvalue corrected for testing all genes (FDR)

Love et al. 2014, Genome Biol 15:550

# Binding and expression together

**binding: sites + closest TSS (toy)**

| site | gene | distance | log2FC | FDR | group |
|---|---|---|---|---|---|
| s1 | geneA | 120 | 1.8 | 0.001 | gained |
| s2 | geneB | 450 | 2.1 | 0.010 | gained |
| s3 | geneB | 800 | 0.2 | 0.600 | (see s2) |
| s4 | geneC | 300 | 0.1 | 0.700 | unchanged |
| s5 | geneD | 2,400 | 1.5 | 0.002 | too far |
| s6 | geneE | 90 | -1.2 | 0.020 | lost: none |
| s7 | geneF | 600 | 0.3 | 0.400 | unchanged |
| s8 | geneG | 50 | 1.1 | 0.030 | gained |

keep distance ≤ 1,000
gained: FDR < 0.05 and log2FC > 0
unchanged: FDR ≥ 0.05, gene not gained

**expression: DESeq2 (toy)**

| gene | log2FC | padj | class |
|---|---|---|---|
| geneA | 2.5 | 0.001 | induced |
| geneB | 0.4 | 0.300 | n.s. |
| geneG | -1.1 | 0.010 | repressed |
| geneC | 1.2 | 0.020 | induced |
| geneF | 0.1 | 0.900 | n.s. |

+ 7 other genes

induced: padj < 0.05, log2FC > 0
repressed: padj < 0.05, log2FC < 0

**Is the share of induced genes higher with gained binding?**
→ your data, Part 3.3

**% of genes (toy)**

